Updated: April 6, 2026

SSR "GNSS Forever Board©'' – *1024 Week Roll-Over Free

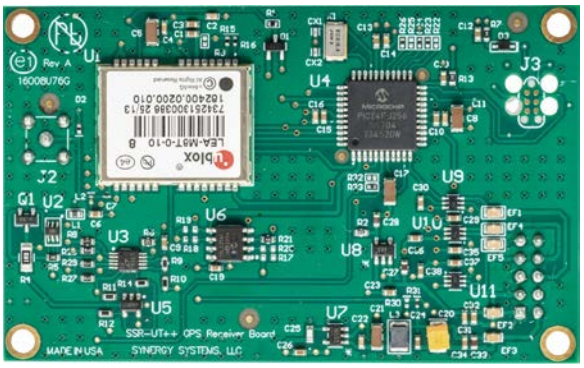
SSR-M8T++ DROP-IN** REPLACEMENTS FOR MOTOROLA 6 & 8 Channel VP, 8 Channel UT+ & GT+ RECEIVERS

Also Replaces smaller formfactor 12 Channel Motorola and iLotus Receivers Mounted on larger VP/UT+ Size Adaptor Boards

UPDATES OLDER GPS BASED NAVIGATION AND TIMING INSTRUMENTATION PRODUCTS

Introduction:

SSR-M8T++ Navigation and Precision Timing receivers embed a state-of-the-art u-Blox 72 channel, LEA-M8T Multi-GNSS modules. SSR Series replacements are form, fit, and plug-compatible with the original Motorola VP, GT+ and UT+ OEM GPS receiver boards and include the most popular binary messages and content.



Features:

- **Designed for Navigation and timing applications**
- **Motorola binary message emulation at 9600 baud for backward compatibility with legacy Motorola 6 and 8 channel VP and 8 channel UT+ and GT+ receivers.**
- **12 channels included for updated UT+ legacy products**
- **Super-Fast TTFF and State-of-the-Art Sensitivity**
- **More satellites tracked than receivers being replaced**
- **Better ADEV performance in legacy timing instruments**
- **Drop-In to most VP, GT+ and UT+ GPS receiver slots Without the need for user intervention**
- ***No Receiver 1024 Week Roll-Overs!**
- **Set to GPS only mode for backward compatibility**

Physical Characteristics:

SSR-M8T++ receivers use Motorola's VP, UT+ and GT+ Oncore, and Synergy Adaptor Board, industry standard, 2.0 in x 3.25 in size, mounting hole locations, connector type and positions.

Electrical Characteristics:

The high performance SSR-M8T++ boards embed popular u-Blox LEA-M8T Flash based GNSS modules operating at 2.75 to 3.5 VDC. TxD, RxD communications and Input voltage are converted to 5VDC for compatibility with Motorola's legacy GPS receiver boards.

SSR Series receivers feature an on-board microprocessor to generate emulated 6 and 8 channel Motorola binary messages. An Antenna Power Management System with antenna under current alert and over current protection is designed to supply 5VDC at up to 90 mA to a wide range of GPS antennas. "Tag-Connect" re-programming port included for potential future firmware updates.

* 1024 Week Roll-Over Free* with firmware versions 4.xx and above.
** For "Drop-In" Compatibility, and previous versions, see page 2.

PHYSICAL CONSTRUCTION	
Dimensions	50.8 mm x 82.55 mm x 5 mm (2.0 in x 3.25 in x 0.5 in)
Weight	16 grams
Data/Power Header	10 Pin, 2x5 header, 2.54 mm (0.10 in) pitch
Antenna Connector	MCX

OPERATIONAL CHARACTERISTICS	
Architecture	72 channels with over 2 million correlators
L1 GNSS Supported	GPS, QZSS, GLONASS, BeiDou, Galileo
SBAS Supported	WAAS, EGNOS, MSAS, GAGAN, KASS
Tracking Channels	72 channels **(6, 8 and 12 channels for Motorola compatibility)**
Acquisition Times:	
Hot Start	< 1s Typical for all GNSS
Cold Start	GPS 29s
Position Accuracy	<2.5 m Autonomous - <2.0 m SBAS - <2.0 m RTCM-104
Heading Accuracy	0.3° Typical for GPS
Sensitivity	-148 dBm at cold start -167 dBm while tracking (-160 dBm Reacquisition)
Power Supply	5 VDC +/- 0.25 V 0.45 mA at 5.0 V Typical
Backup Power	+1.4 to +3.6V at 22 uA Max
Temperature Range	-40 Degrees to + 85 Degrees
Storage Temperature	-40 Degrees to + 85 Degrees
Humidity	95% over dry bulb range of +38°C to +85°C

1PPS TIMING PULSE	
Accuracy of 1PPS	≤ 20 ns, <60 ns 99%
Compensated	<15 ns (quantization error applied)
Granularity	21 ns

COMMUNICATIONS INTERFACE DEFAULT	
Default Protocol	Motorola 6, 8 and 12 channel at 9600 baud
Default Status	1PPS output active at power-on, no message outputs
Update Rate	1Hz

Request Synergy's "VP Commands Reference" document for Motorola binary message details. For 8 channel UT+ and GT+ message details, refer to the GT_UTMAN User's Guide: https://synergy-gps.com/motorola/. Download WinOncore12 software in the Support section of Synergy's web site listed below, or Tac32PLus here: https://gpstime.com/cns-systems-inc/tac32plus_sw/. Request Tech-Note #498 for 6, 8 and 12 channel Command-Reply Message availability. For Remote Re-Programming select an appropriate Tag-Connect cable here: https://www.tag-connect.com/product-category/products/cables

Ordering Information - See pages 5, 6 & 7 for VP, UT+ & GT+ Clone Part Numbers

Part Number	Full SSR-M8T++ Receiver Configuration
16UT4726G-2	Multi-GNSS - Mot Binary emulation; UBX NMEA, Binary; R/A MCX RF Connector
16UT4724G-1	Multi-GNSS - Mot Binary emulation; UBX NMEA, Binary; Straight MCX RF Connector

Ordering Information for replacing 12 Channel Receivers on Adaptor Boards

Part Number	Replaces	SSR-M8T++ Receiver Configuration Description
16UT4726G-P2	Motorola M12+	12 Channel Motorola Binary; R/A MCX Connector
16UT4724G-P2	Motorola M12+	12 Channel Motorola Binary; Straight MCX Connector
16UT4726G-P3	iLotus M12M	12 Channel Motorola Binary; R/A MCX Connector
16UT4724G-P3	iLotus M12M	12 Channel Motorola Binary; Straight MCX Connector

Note: See page 6 for compatibility with precision timing instrumentation products

For configuration assistance, order placement and technical support call or Email:

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Web: www.synergy-gps.com

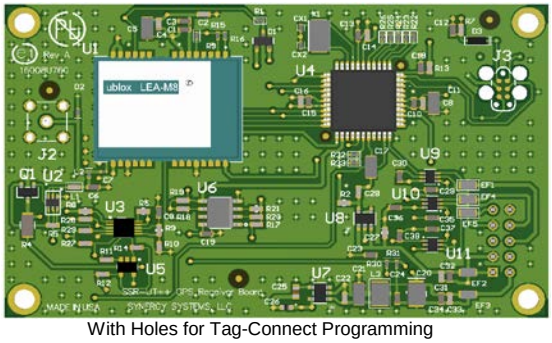
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DROP-IN* REPLACEMENT BOARDS FOR MOTOROLA VP, GT+ and UT+ OEM RECEIVERS

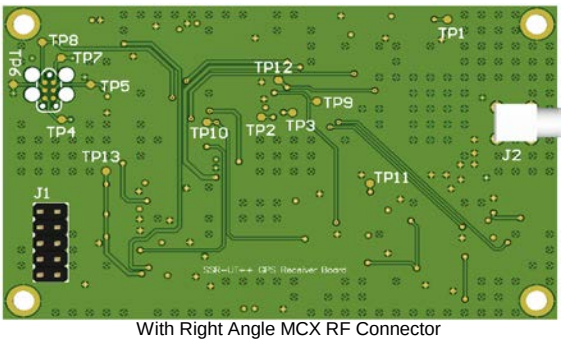
SSR-M8T++ - Large formfactor board layout and connector position detail

Component Layout - Top



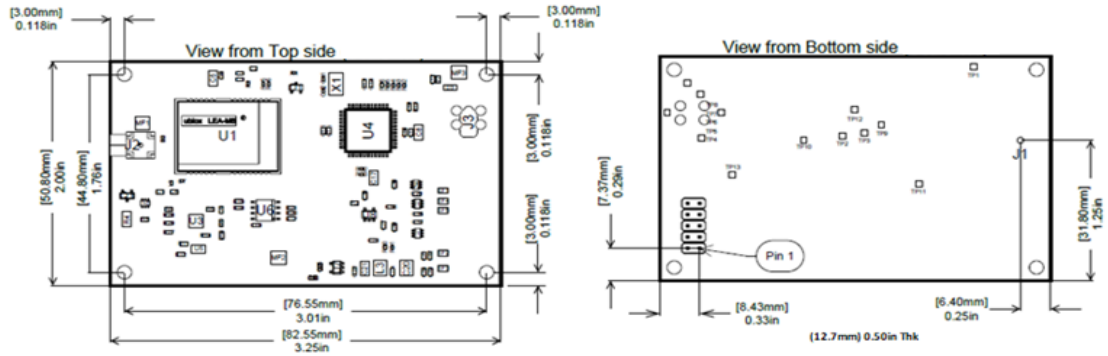
With Holes for Tag-Connect Programming

Connector Layout - Bottom



With Right Angle MCX RF Connector

SSR-M8T++ Board Dimensions



10 pin header I/O pinouts

Pin #	
1	externally applied backup power
2	+5 VDC main power
3	ground
4	N/A
5	N/A
6	1PPS
7	1PPS return
8	TTL TxD messages from receiver
9	TTL RxD commands to receiver
10	TTL Return

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DROP-IN* REPLACEMENT BOARDS FOR MOTOROLA VP, GT+ and UT+ OEM RECEIVERS

SSR-M8T++ Design Comments

1. “Drop-In” Compatibility** - SSR Series receivers include emulation of the most popular Motorola binary command and reply messages that Synergy supported over 27 years as a distributor for Motorola and iLotus OEM GPS receiver products. Some reply messages may not have 100% of the binary bits listed in the Motorola's VP, GT+ and UT+ User's Guides: https://synergy-gps.com/support/. That's due to not being needed by Synergy's customer base or raw data was not available in the more modern, updated u-Blox receiver architecture. Messages, either standard or custom may be added (minimum quantity or NRE may apply).

2. u-Blox based SSR receivers use a modern receiver architecture, completely different from Motorola's series of receivers with a firmware base originally designed in the late nineteen eighties (Eagle IV). It is impractical to emulate an exact duplicate of the Motorola binary protocol and it is not a compatibility goal. An exact duplicate could actually limit the performance advancements that u-Blox modules offer.

3. SSR-M8T++ receivers include a "J3" PCB layout for the "Tag-Connect," a Connector-less method for re-programming the emulation Microprocessor. (Motorola did not offer user firmware updates). For SSR-M8T++ receivers, a Microchip ICD, PICkit, or similar in-circuit programming tool, can be used to update the processor when firmware updates become available. See Tag-Connect's standard programming cables for Microchip's ICD and PICkit series programmers here: https://www.tag-connect.com/product-category/products/cables

4. Unlike previous Motorola receivers, a backup battery is not required to hold an Almanac and last position to produce a faster Time to First Fix (TTFF). An external backup voltage is recommended if running the Real Time Clock (RTC) if required while the receiver is powered off. An optional (special order) SSR part number is available with an on-board 11 mAh rechargeable battery to operate the RTC for about 2 weeks when the receiver is powered down. The on-board battery reduces the normal operating and storage temperature range from -40°c to +85°c to -20°c to +60°c.

5. Refer to u-Blox LEA-M8T GNSS module documents for full module technical and performance specifications here: https://www.ubl https://www.ublox.com

6. For reference to previous SSR-6Tf + and SSR-M8T+ models with 128K emulation processors, see Synergy Tech-Note #484

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Features and Specification Comparison
Legacy Motorola VP and UT+ vs. SSR-VP/UT+ Clone Receivers using the updated SSR-M8T++

Feature	Motorola 6 and 8 channel VP Oncore	Motorola 8 channel UT+ Oncore	Synergy SSR-M8T+
Susceptibility to 1024 Week Roll-Over	All models have already rolled over	All models have already rolled over	**NO 1024 Week Roll-Overs** **GPS Forever Board©**
Tracking Capability	• 6 or 8 parallel channels • L1 1575.42 MHz • C/A code (1.023 MHz chip rate)	• 6 or 8 parallel channels • L1 1575.42 MHz • C/A code (1.023 MHz chip rate)	• 72 channels, reduced to 6, 8 and 12 channel "Best in View" to match legacy Motorola channel number requirements • L1 1575.42 MHz • C/A code (1.023 MHz chip rate)
SBAS	No	No	• WAAS, EGNOS, MSAS
Acquisition Time (Time To First Fix)	• cold - 180 s • Warm - 45 s typical • Hot - 20 s typical • Reacquisition - 2.5s	• cold - 300 s typical • Warm - 50 s typical • Hot - 20 s typical • Reacquisition - < 1 s	• cold - 26 s • Warm - < 2 s typical • Hot - 1 s • Reacquisition – < 1 s
Precision Timing	• 1 PPS timing output with Time RAIM	Same	Same
Timing Accuracy	• < 50 ns observed - position hold	• < 50 ns observed - position hold	• < 20 ns - position hold
Output Messages	• Latitude, longitude, height, velocity, heading, time, satellite tracking status (Motorola binary protocol) • Software selectable output rate (Continuous or poll) • Broad list of commands/messages • TTL interface	• Latitude, longitude, height, velocity, heading, time, satellite tracking status (Motorola binary protocol) • Software selectable output rate (Continuous or poll) • Broad list of commands/messages • TTL interface	• Latitude, longitude, height, velocity, heading, time, satellite tracking status (Emulated Motorola binary protocol) • Software selectable output rate (Continuous or poll) • Popular timing commands/messages • Broad list of timing commands/messages • TTL interface
Programmable Processor	No	No	Yes (To add features and/or future commands)
Power Requirements	• 5 ± 0.25 Vdc; 50 mV p-p ripple (max)	• Same	•Same
Power Consumption	• 5 VDC, 1.1 W	• 5 VDC, 0.9 W	• 5 VDC, 0.5W
Antenna Drive	• 5 VDC Powered through receiver module	• Same	•More robust Antenna 3V-5V Power Management System
Dimensions	• 2.00 x 3.25 x 0.64 in. [50.8 x 82.6 x 16.3 mm]	• Same	• Same
Weight	• 1.8 oz. (51 g)	• same	• 22 Grams
Connectors	• Data/power: 10 pin (2x5) header, 0.10" centers • RF: right angle OSX (subminiature Snap-On)	•Same • Same	• Same • MCX Backward Compatible with OSX
Operating Temperature	• -30ºC to +85ºC	• -40ºC to +85ºC	• -40ºC to +85ºC
Operating Period	• 1024 Weeks	• 1024 Weeks	• Un-Limited
Humidity	• 95% noncondensing +30ºC to +60ºC	• Same	• Same
Certifications	• FCC	• Same	• FCC, Industry Canada, CE (tested in SynPaQ®/E)

Note: GT+ Oncore navigation receiver the same as the UT+ but with 500 ns PPS and without timing features like T-RAIM and Position Hold

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SSR SERIES DROP-IN REPLACEMENTS FOR SYNERGY'S ORIGINAL 1000xxx Series M12+ And M12M, 12 CHANNEL ADAPTOR BOARDS With New Part Numbers for SSR-M8T++ Receivers

OLD SYNERGY ADAPTOR BOARD to be REPLACED		SYNERGY'S NEW SINGLE-BOARD REPLACEMENT ORDERING INFORMATION		
OLD SYNERGY P/N	**OLD MODEL/DESCRIPTION**	**SYNERGY MODEL**	**NEW DESCRIPTION**	**SYNERGY P/N**
10001872 w/ M12M	M12M-T 12 channel receiver on Adaptor Board, no bat, Straight 10 pin I/O header, R/A MCX Jack	SSR-M12M-T	SSR-M8T+, 12 channel, Single Board receiver, no bat, Straight 10 pin I/O header, R/A MCX Jack	16UT4726G-P3
10001472 w/ M12+	M12+/ T 12 channel receiver on Adaptor Board, no bat, Straight 10 pin I/O header, R/A MCX Jack	SSR-M12+	SSR-M8T+, 12 channel, Single Board receiver, no bat, Straight 10 pin I/O header, R/A MCX Jack	16UT4726G-P2
10001874 w/ M12M	M12M-T 12 channel receiver on Adaptor Board, no bat, Straight 10 pin I/O header, Str MCX Jack	SSR-M12M-T/S	SSR-M8T, 12 channel, Single Board receiver, no bat, Straight 10 pin I/O header, straight MCX Jack	16UT4724G-P3
10001474 w/ M12+	M12+/T 12 channel receiver on Adaptor Board, no bat, Straight 10 pin I/O header, Str MCX Jack	SSR-M12+/S	SSR-M8T, 12 channel, Single Board receiver, no bat, Straight 10 pin I/O header, straight MCX Jack	16UT4724G-P2
10001827	M12M-T 12 channel Shielded receiver on Adaptor Board, no bat, Straight 10 pin I/O header, R/A MCX Jack	SSR-M12M-T	SSR-M8T, 12 channel, Single Board receiver, no bat, Straight 10 pin I/O header, R/A MCX Jack, No extra shield required	16UT4726G-P3
10001860G	M12M-N 12 channel Navigation receiver on Adaptor Board, W/Bat, Straight 10 pin I/O header, R/A MCX Jack	SSR-M12M-N	SSR-M8T, 12 channel, Single Board receiver, No Bat, Straight 10 pin I/O header, R/A MCX Jack (needs user supplied battery B/U on pin 1 of header)	16UT4726G-P3
10001729 Motorola P/N MOTO187995D01	M12M-T 12 channel Shielded receiver on Adaptor Board, no bat, Straight 10 pin I/O header, R/A MCX Jack	SSR-M12M-T	SSR-M8T, 12 channel, Single Board receiver, no bat, Straight 10 pin I/O header, R/A MCX Jack, No extra shield required	16UT4726G-P3

Notes:

1. The Synergy Model Numbers shown in this table are designed to work in Motorola, and other manufacturer's products where the product's internal software has been updated from the original 6 or 8 channel Motorola binary code to recognize 12 channel binary command\reply messages produced by Motorola's original M12+ and follow-on iLotus M12M GPS receivers.
2. The original Synergy Adaptor Board product P/N 10001729 and P/N 10001827 with an overall EMI Shield can now be replaced with the P/N 16UT4525G-P3 since the entire GPS module on the SSR-M12M-T is shielded - an additional overall shield is not required.
3. SSR Series timing receivers offer excellent tracking, positioning and navigation results. For navigation applications, please send the "platform maximum expected acceleration and velocity" to Synergy so the SSR receiver dynamics can be set prior to shipment.
4. Adaptor Board P/N 10001860G included an on-board keep-alive battery to hold Almanac, position and time (including an RTC) in RAM to aid in a faster Time to First Fix (TTFF). The replacement SSR-M12T features much faster TTFF so a backup battery is no longer necessary. Non-battery SSR boards also eliminate the need to replace the battery in the field at the end of their service life (8 – 12 years depending on environmental factors).
5. For timing products using non-Synergy Adaptor Boards, please contact Synergy at oemtech@synergy-gps.com.

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Manufacturers will be added to this page as their products are tested April 6, 2026

SSR-M8T++ Receiver Selection chart and Ordering Instructions

Drop-In Replacements for Legacy Motorola 6 and 8 Channel VP and 8 Channel UT+ and GT+ Oncore GPS Receivers

10 Pin Header and MCX RF Connector are both bottom mounted for backward compatibility with Motorola boards

Legacy timing products were designed for GPS only operation. Multi-GNSS operation is incompatible and not offered or supported

MOTOROLA PRODUCT to be REPLACED			SYNERGY REPLACEMENT		MANUFACTURER'S TIMING PRODUCT
Motorola Model	**Motorola P/N**	**Receiver Description**	**Synergy Clone Model #**	**Synergy P/N**	**Timing Instrument Manufacturers**
VP-B1 6 Channel	B1xxxXxxxX	6 channel VP Oncore – R/A OSX	SSR-VP1/R	16UT4726G-B1	Austron\Datum System2000***; HP Z3801A and Z3805A ; HP 59551A
VP-B3 8 Channel	B3xxxXxxxX	8 channel VP Oncore – R/A OSX	SSR-VP3/R	16UT4726G-B3	
VP-B4 8 Channel	B4xxxXxxxX	8 channel VP Oncore – R/A OSX	SSR-VP4/R	16UT4726G-B4	
VP-B4 8 Channel	B4xxxXxx5X	8 channel VP Oncore – STR OSX	SSR-VP4/S	16UT4724G-B4	HP Z3816A; Lucent KS-24361 REF-1****
VP-B8 8 Channel	B8xxxXxx1X	8 channel VP Oncore – R/A OSX	SSR-VP8/R	16UT4726G-B8	HP 58503A; HP 58503B*; Symmetricom SSU-2000 ***
VP-B8 8 Channel	B8xxxXxx5X	8 channel VP Oncore – STR OSX	SSR-VP8/S	16UT4724G-B8	
UT+ 8 Channel	R5xxxUxxx1X	8 channel UT+ Oncore – R/A OSX	SSR-UT+/R	16UT4726G-R5	Symmetricom SSU 2000 ; HP 58503B* Datum/Symmetricom ET6000**; Zypher GPStarPlus** HP Z3811; Lucent KS-24361****
UT+ 8 Channel	R5xxxUxxx5X	8 channel UT+ Oncore – STR OSX UT+ size 12 Channel M12+ Clone	SSR-UT+/S	16UT4724G-R5	Zypher CommSync II; HP Z3811; Lucent KS-24361**** HP 58503B*; TRAK Micro 9001-1 GPS cards. Efratom Model 137, 16 Motorola Simulcast
GT+ 8 Channel	R3xxxUxxx1X	8 channel GT+ Oncore – R/A OSX	SSR-GT+/R	16UT4726G-R3	General Navigation and for +/- 500 ns Timing Instruments

* Original HP 58503B units were delivered with 8 channel Motorola VP Oncore receivers. Later units used an 8 channel UT+ Oncore with either a right angle MCX or a straight MCX RF connector. Some were delivered with 12 channel iLotus M12M receivers on non-Synergy Adaptor Boards that output 8 channel messages. See Note B below.

** Early GPStar and ET6000 products were produced with Motorola 8 channel UT+ Oncore receivers. Later products were shipped with a smaller formfactor 12 channel Motorola M12+ mounted on a Synergy Adaptor Board P/N 10001874 that would plug into the larger formfactor UT+ slot. This SSR replacement board can be set for 8 or 12 channel operation at the factory (photo required prior to placing an order).

*** Early SSU-2000 units were produced with Motorola 6 channel VP Oncore receivers. Next, they were shipped with 8 channel VP and UT+ Oncore receivers and finally 12 channel Motorola M12+ receivers mounted on a Synergy Adaptor Board that would plug into the larger formfactor VP/UT+ slot. iLotus 12 channel M12M receivers were the last of the Motorola designed receivers to be used. This SSR replacement board can be set for 8 or 12 channel operation at the factory (photo required prior to placing an order).

**** Early Lucent KS-24361 Ref-1 products were produced with Motorola 8 channel VP receivers. Later products were shipped Motorola 8 channel UT+ receivers. Some shipped with R/A MCX and others with Straight MCX

To Order: A. Determine a part number from pages 4 and 5 or a manufacturers name and product model number from this page (for example HP Model Z3801A). If a Manufacturer's name and product model is not included on this page, please e-mail a front panel photo of the manufacturer's product and the top and bottom of the existing GPS board.

B. E-mail a photo of the top and bottom of the Motorola board (and, if used, the bottom of an Adaptor Board's RF connector). Some timing products manufacturers switched from right angle to straight MCX RF connectors during their production phase and the proper SSR part number needs to be specified.

C. Synergy will return confirmation of the replacement SSR receiver part number from page 4, 5 or 6 and quote a price for the number of receivers desired.

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April 6, 2026

SSR-M8T++ Replacement Receiver Selection chart for updating Lucent Technologies Legacy Products Designed for Motorola VP and UT+ Oncore OEM boards

Lucent Technologies Product	Original GPS Receiver Type	Channels	Synergy Replacement Model	Synergy P/N
SII - RFTG L104B/C, L106A/B	Motorola - VP Oncore B1	6	SSR-VP1/R	16UT4726G-B1
SII - RFTG L106C, L109	Motorola VP Oncore B3	8	SSR-VP3/R	16UT4726G-B3 *
SII - RFTG L106C, L109	Motorola VP Oncore B4	8	SSR-VP4/R	16UT4726G-B4 *
SII - RFTG L106C, L109	Motorola VP Oncore B4	8	SSR-VP4/S	16UT4724G-B4 *
SII - RFTG L106C, L109	Motorola VP Oncore B8	8	SSR-VP8/R	16UT4726G-B8 *
KS-24361 REF-0 REF-1	Motorola UT+ Oncore R5	8	SSR-UT+/S	16UT4724G-R5 **
SII - RFTGu	Motorola UT + Oncore R5	8	SSR-UT+/R	16UT4726G-R5
RFTGM-II XO	Motorola UT+ Oncore R5	8	SSR-UT+/R	16UT4726G-R5 **
RFTGM-II XO	Motorola UT+ Oncore R5	8	SSR-UT+/S	16UT4724G-R5 **

* Some SII - RFTG L106C, L109 units may have embedded the VP-B3 or VP-B4 receiver versions prior to migrating to the VP-B8.

** The KS-24361, and some of the RFTGM-II XO products, were manufactured with UT+ receivers with a straight MCX RF connector (/S) and others were shipped with right angle MCX RF connectors (/R). To ensure compatibility, please confirm the manufacturer of the product being updated and send a photo of the top and bottom of the Motorola GPS board to Synergy showing the RF Connector. Synergy will return confirmation of the replacement SSR receiver part number and quote a price for the number of receivers desired.

Note: As a result of a contract agreement with another customer, Synergy does not offer replacement GPS boards for the following Lucent Technologies legacy products: TFU - 44ww7, TFU - 44ww8, TFU - 44ww7B, TFU - 44ww8B, CTU - 1:1, 1:2, CTU - 1:3 and later

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VP, UT+ and GT+ Board Configuration Details

Motorola VP, UT+ and GT+ Oncore OEM GPS precision timing boards were supplied with either a Right Angle or Straight MCX RF connector. Components were populated on both the top and the bottom sides of these boards. Synergy supplies SSR Series receivers with both antenna connector options and duplicates Motorola's VP, UT+ and GT+ receiver's larger board dimensions of 52.80 mm x 82.60 mm (2.00 in x 3.25).



Motorola VP Oncore board with a right angle MCX (UT+ and GT+ are the same)



Motorola VP Oncore board with a straight (STR) MCX (UT+ is the same)

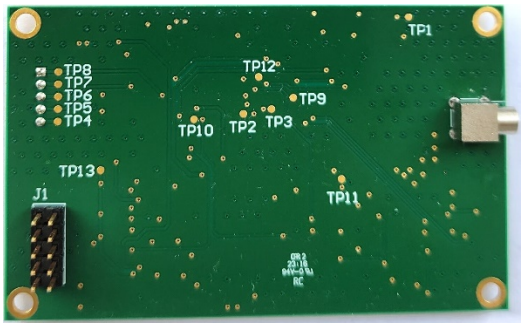
The Straight MCX version allowed both 10 Pin I/O and the RF connector to be simultaneously plugged directly into a motherboard. The UT+ and GT+ RF Connector positions are identical.

The Straight MCX connector on VP and UT+ Oncore receivers allowed designers to include additional antenna protection components on the motherboard. It also eliminated the cost of an RF coax cable. (GT+ Oncore used Right angle MCX only).

GT+ Oncore receivers were used in some lower cost Motorola timing products with ~500 ns timing accuracy and no precision timing features like T-RAIM.

SSR Board Connector Locations

(all components mounted on the top side)

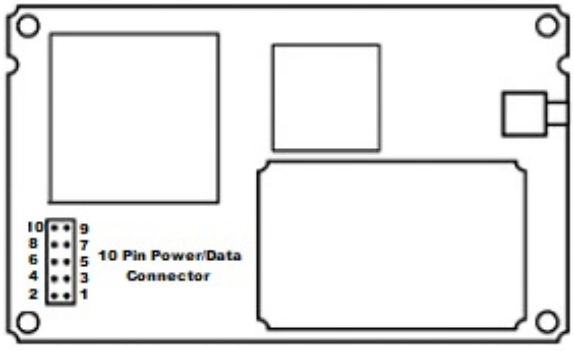


To comply with Motorola VP, UT+ and GT+ standard, SSR Series Drop-In replacement GPS receiver boards are populated with the 10 pin I/O and MCX RF connectors mounted on the bottom side of the board (top mount MCX part numbers quoted on request).

6 & 8 Channel VP, GT+, UT+ and SSR Pinouts

PIN	FUNCTION	DESCRIPTION
1	Battery	External Back-Up
2	Power	5 VDC Main Power
3	GND	Ground
4	VPP	Reprogramming Voltage
5	N/A	Not Used (Do Not Connect)
6	1PPS	Timing Pulse
7	1PPS RTN	Timing Pulse Return (GND)
8	TTL TxD	Receiver Command Input
9	TTL RxD	Receiver output Messages
10	TTL RTN	TTL I/O Return (Ground)

I/O Pinout Locations on VP, UT+ and GT+ Boards



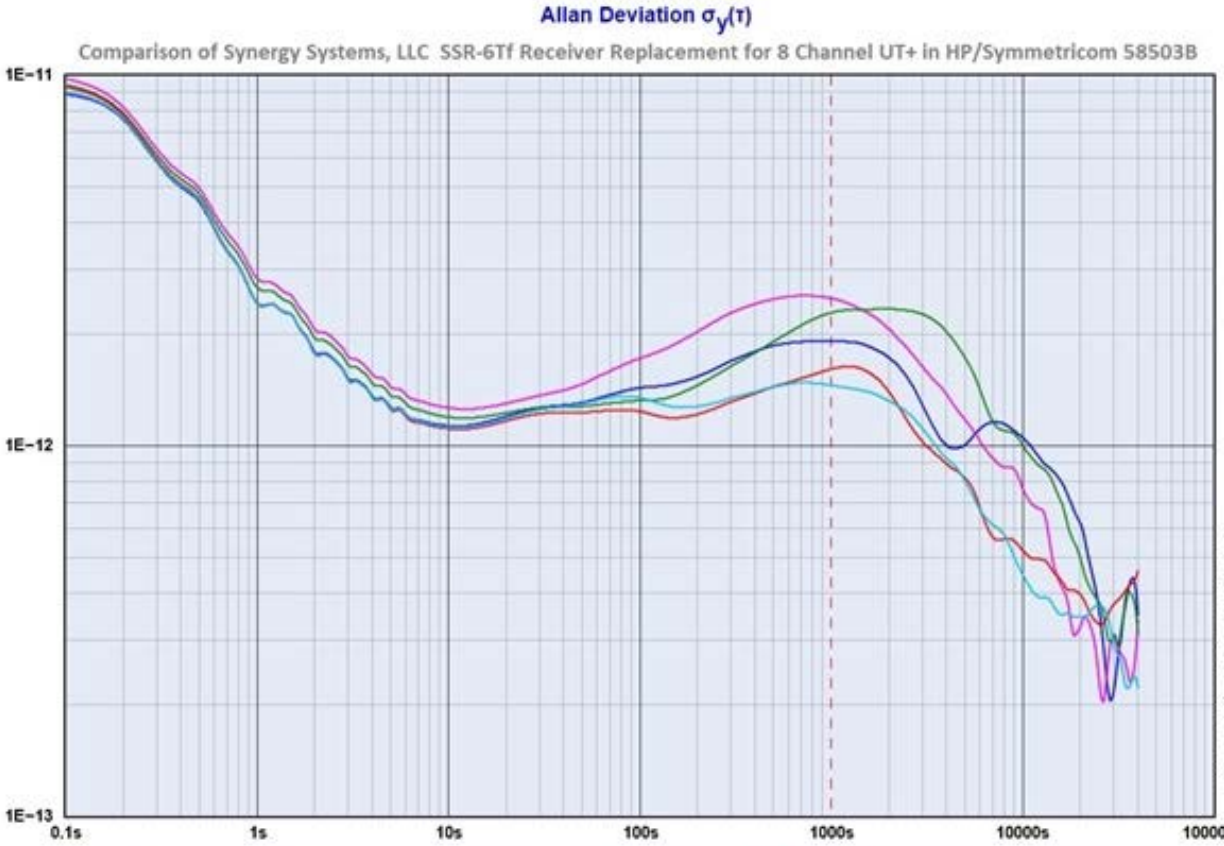
Connector side of the board.

Optional "full enclosure" EMI Shield part number and price are quoted on request.

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Incorporated Test - April 6, 2026

27 October 2023



Tau	Sigma(Tau)
1s	2.42E-12
2s	1.77E-12
4s	1.35E-12
8s	1.15E-12
10s	1.13E-12
20s	1.19E-12
40s	1.28E-12
80s	1.36E-12
100s	1.35E-12
200s	1.28E-12
400s	1.41E-12
800s	1.48E-12
1000s	1.46E-12
2000s	1.32E-12
4000s	9.23E-13
8000s	5.72E-13
10000s	4.46E-13
20000s	3.48E-13
40000s	2.23E-13

Configurations Compared:

Motorola 8 Channel VP-88 Oncore FW v10.0

Motorola 8 Channel UT+ Oncore FW v3.2

Synergy 16 Channel SSR-VP/UT+ Configured as an 8 Channel SSR-UT+ for direct install.

iLotus 12 Channel M12M Mounted to a Goshay Group Adaptor Board Configured to output 8 Channels

Synergy 16 Channel SSR-M12M Clone, no 1024 Roll-Over, configured as a 12 Channel M12M Clone, Mounted to the Goshay Group adaptor board (for conversion to 8 channels for compatibility with the 59503B).

Trace	Input Freq	Sample Interval	ADEV at 1000s	Duration	Acquired	Instrument
53503B with VP 10.0	10.0 MHz	0.100 s	1.92E-12	1d 0h 0m 0s	864000 pts	Phase Station 53100A
53503B with UT+ V3.2	10.0 MHz	0.100 s	2.51E-12	1d 0h 0m 0s	864000 pts	Phase Station 53100A
53503B with Goshay + M12+	10.0 MHz	0.100 s	2.30E-12	1d 0h 0m 0s	864000 pts	Phase Station 53100A
53503B with Goshay + SSR-6Tf+	10.0 MHz	0.100 s	1.62E-12	1d 0h 0m 0s	864000 pts	Phase Station 53100A
53503B with SSR-VP/UT+	10.0 MHz	0.100 s	1.46E-12	1 d	857751 pts	Phase Station 53100A

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